



Unit-4

Partial Derivatives



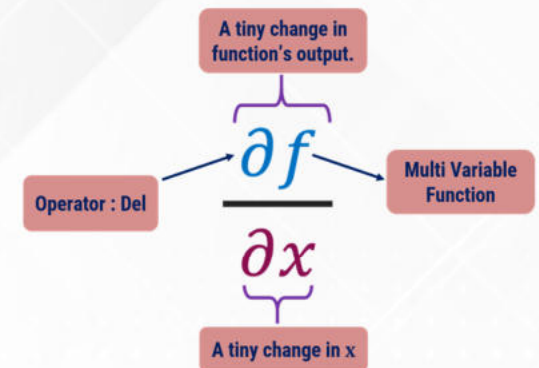
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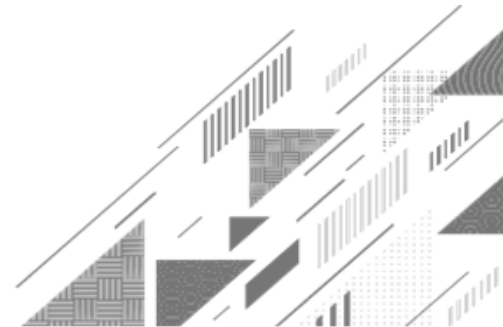
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Method:3 ➡ Partial Derivatives



Partial Derivatives

► Let $f(x, y)$ be a function of two independent variables x and y .

The first order partial derivative of f with respect to x is denoted by

$$f_x \text{ OR } \frac{\partial f}{\partial x} \text{ OR } \left(\frac{\partial f}{\partial x} \right)_{y \text{ is constant}}$$

and it is defined as

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}$$

The first order partial derivative of f with respect to y is denoted by

$$f_y \text{ OR } \frac{\partial f}{\partial y} \text{ OR } \left(\frac{\partial f}{\partial y} \right)_{x \text{ is constant}}$$

and it is defined as

$$\frac{\partial f}{\partial y} = \lim_{k \rightarrow 0} \frac{f(x, y + k) - f(x, y)}{k}$$

Example of First Order Partial Derivatives

Example: If $f(x, y) = x^2 - 3xy + y^2 - 3$ then find $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y}$.

Solution:

$$\begin{aligned}\rightarrow \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x} (x^2 - 3xy + y^2 - 3) \\ &= \frac{\partial}{\partial x} (x^2) - \frac{\partial}{\partial x} (3xy) + \frac{\partial}{\partial x} (y^2) - \frac{\partial}{\partial x} (3) \\ &= 2x - 3y(1) + 0 - 0\end{aligned}$$

$$\Rightarrow \frac{\partial f}{\partial x} = 2x - 3y$$

Example of First Order Partial Derivatives

Example: If $f(x, y) = x^2 - 3xy + y^2 - 3$ then find $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y}$.

Solution Continue:

$$\begin{aligned}\rightarrow \frac{\partial f}{\partial y} &= \frac{\partial}{\partial y} (\quad) \\ &= \frac{\partial}{\partial y} (x^2) - \frac{\partial}{\partial y} (3x y) + \frac{\partial}{\partial y} (y^2) - \frac{\partial}{\partial y} (3) \\ &= 0 - 3x(1) + 2y + 0\end{aligned}$$

$$\Rightarrow \boxed{\frac{\partial f}{\partial y} = 2y - 3x}$$

Example of First Order Partial Derivatives by Definition

Example 1: If $f(x, y) = x^2y + xy^2$, then find $f_x(1, 2)$ & $f_y(1, 2)$ by definition.

Solution:

$$\begin{aligned}\rightarrow f_x &= \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \\&= \lim_{h \rightarrow 0} \frac{(x+h)^2y + (x+h)y^2 - (x^2y + xy^2)}{h} \\&= \lim_{h \rightarrow 0} \frac{(x^2 + 2hx + h^2)y + (x+h)y^2 - x^2y - xy^2}{h} \\&= \lim_{h \rightarrow 0} \frac{\cancel{x^2y} + 2hxy + h^2y + \cancel{xy^2} + hy^2 - \cancel{x^2y} - \cancel{xy^2}}{h}\end{aligned}$$

Example of First Order Partial Derivatives by Definition

Example 1: If $f(x, y) = x^2y + xy^2$, then find $f_x(1, 2)$ & $f_y(1, 2)$ by definition.

Solution Continue:

$$= \lim_{h \rightarrow 0} \frac{2hxy + h^2y + hy^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h}(2xy + hy + y^2)}{\cancel{h}}$$

$$= \lim_{h \rightarrow 0} 2xy + hy + y^2$$

$$= 2xy + 0y + y^2$$

$$\Rightarrow \boxed{f_x = 2xy + y^2}$$

$$\Rightarrow f_x(1, 2) = 2(1)(2) + (2)^2$$

$$\Rightarrow \boxed{f_x(1, 2) = 8}$$

Example of First Order Partial Derivatives by Definition

Example 1: If $f(x, y) = x^2y + xy^2$, then find $f_x(1, 2)$ & $f_y(1, 2)$ by definition.

Solution Continue:

$$\begin{aligned}\rightarrow f_y &= \lim_{k \rightarrow 0} \frac{f(x, y+k) - f(x, y)}{k} \\&= \lim_{k \rightarrow 0} \frac{x^2(y+k) + x(y+k)^2 - (x^2y + xy^2)}{k} \\&= \lim_{k \rightarrow 0} \frac{x^2y + kx^2 + x(y^2 + 2ky + k^2) - x^2y - xy^2}{k} \\&= \lim_{k \rightarrow 0} \frac{\cancel{x^2y} + kx^2 + \cancel{xy^2} + 2kxy + k^2x - \cancel{x^2y} - \cancel{xy^2}}{k}\end{aligned}$$

Example of First Order Partial Derivatives by Definition

Example 1: If $f(x, y) = x^2y + xy^2$, then find $f_x(1, 2)$ & $f_y(1, 2)$ by definition.

Solution Continue:

$$= \lim_{k \rightarrow 0} \frac{kx^2 + 2kxy + k^2x}{k}$$

$$= \lim_{k \rightarrow 0} \frac{\cancel{k}(x^2 + 2xy + kx)}{\cancel{k}}$$

$$= \lim_{k \rightarrow 0} x^2 + 2xy + kx$$

$$= x^2 + 2xy$$

$$\Rightarrow \boxed{f_y = x^2 + 2xy}$$

$$\Rightarrow f_y(1, 2) = (1)^2 + 2(1)(2)$$

$$\Rightarrow \boxed{f_y(1, 2) = 5}$$

Example of Partial Derivatives

Example 3: If $f(x, y) = x^3 + x^2y^3 - 2y^2$, find $f_x(2, 1)$ & $f_y(2, 1)$.

Solution:

Here, $f(x, y) = x^3 + x^2y^3 - 2y^2$.

$$\begin{aligned}\Rightarrow f_x &= \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (x^3 + x^2y^3 - 2y^2) \\ &= \frac{\partial}{\partial x} (x^3) + \frac{\partial}{\partial x} (x^2y^3) - \frac{\partial}{\partial x} (2y^2) \\ &= 3x^2 + 2xy^3 - 0 \\ \Rightarrow f_x &= 3x^2 + 2xy^3\end{aligned}$$

$$\begin{aligned}\text{Now, } f_x(2, 1) &= 3(2)^2 + 2(2)(1)^3 \\ &= 12 + 4 \\ \Rightarrow f_x(2, 1) &= 16\end{aligned}$$

Example of Partial Derivatives

Example 3: If $f(x, y) = x^3 + x^2y^3 - 2y^2$, find $f_x(2, 1)$ & $f_y(2, 1)$.

Solution Continue:

Here, $f(x, y) = x^3 + x^2y^3 - 2y^2$.

$$\begin{aligned}\Rightarrow f_y &= \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (x^3 + x^2y^3 - 2y^2) \\ &= \frac{\partial}{\partial y} (x^3) + \frac{\partial}{\partial y} (x^2y^3) - \frac{\partial}{\partial y} (2y^2) \\ &= 0 + 3x^2y^2 - 4y \\ \Rightarrow f_y &= 3x^2y^2 - 4y\end{aligned}$$

$$\begin{aligned}\text{Now, } f_y(2, 1) &= 3(2)^2(1)^2 - 4(1) \\ &= 12 - 4 \\ \Rightarrow f_y(2, 1) &= 8\end{aligned}$$

Example of Partial Derivatives

Example 5: If $f(x, y) = \sin\left(\frac{x}{1+y}\right)$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution:

$$\begin{aligned}\rightarrow f(x, y) &= \sin\left(\frac{x}{1+y}\right) \\ \Rightarrow \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x} \left\{ \sin\left(\frac{x}{1+y}\right) \right\} \\ &= \cos\left(\frac{x}{1+y}\right) \cdot \frac{\partial}{\partial x} \left(\frac{x}{1+y} \right) \\ &= \cos\left(\frac{x}{1+y}\right) \cdot \frac{1}{1+y} \cdot \frac{\partial}{\partial x} (x)\end{aligned}$$

$$\begin{aligned}&= \cos\left(\frac{x}{1+y}\right) \cdot \frac{1}{1+y} \cdot 1 \\ \Rightarrow \frac{\partial f}{\partial x} &= \frac{1}{1+y} \cdot \cos\left(\frac{x}{1+y}\right)\end{aligned}$$

Example of Partial Derivatives

Example 5: If $f(x, y) = \sin\left(\frac{x}{1+y}\right)$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution Continue:

$$\begin{aligned}\rightarrow f(x, y) &= \sin\left(\frac{x}{1+y}\right) \\ \Rightarrow \frac{\partial f}{\partial y} &= \frac{\partial}{\partial y} \left\{ \sin\left(\frac{x}{1+y}\right) \right\} \\ &= \cos\left(\frac{x}{1+y}\right) \cdot \frac{\partial}{\partial y} \left(\frac{x}{1+y} \right) \\ &= \cos\left(\frac{x}{1+y}\right) \cdot x \cdot \frac{\partial}{\partial y} \left(\frac{1}{1+y} \right)\end{aligned}$$

$$\begin{aligned}&= \cos\left(\frac{x}{1+y}\right) \cdot x \cdot \frac{-1}{(1+y)^2} \\ \Rightarrow \frac{\partial f}{\partial y} &= \frac{-x}{(1+y)^2} \cdot \cos\left(\frac{x}{1+y}\right)\end{aligned}$$

Higher Order Partial Derivative

- ▶ Let $f(x, y)$ be a function of two independent variables x and y .

The second order partial derivatives are as follow:

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} = f_{xx}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = f_{xy}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} = f_{yx}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} = f_{yy}$$

- ▶ Mixed Derivative Theorem (Clairaut's Theorem)

If $f(x, y)$, f_x , f_y , f_{xy} & f_{yx} are defined throughout an open region containing a point (a, b) and f_x , f_y , f_{xy} & f_{yx} are continuous at (a, b) then $f_{xy}(a, b) = f_{yx}(a, b)$.

Example of Partial Derivatives

Example 4: Find all second order partial derivative of $f(x, y) = x^3 + x^2y^3 - 2y^2$.

Solution:

$$\begin{aligned} \rightarrow f(x, y) &= x^3 + x^2y^3 - 2y^2 \\ \Rightarrow f_x &= \frac{\partial f}{\partial x} = 3x^2 + 2xy^3 \\ \& f_y &= \frac{\partial f}{\partial y} = 3x^2y^2 - 4y \end{aligned}$$
$$\begin{aligned} f_{xx} &= \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (3x^2 + 2xy^3) = 6x + 2y^3 \\ f_{xy} &= \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (3x^2 + 2xy^3) = 6xy^2 \\ f_{yx} &= \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (3x^2y^2 - 4y) = 6xy^2 \\ f_{yy} &= \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (3x^2y^2 - 4y) = 6x^2y - 4 \end{aligned}$$

Higher Order Partial Derivative

- Let $f(x, y)$ be a function of two independent variables x and y .
The third order partial derivatives are as follow:

$$\frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) \right) = \frac{\partial^3 f}{\partial x^3} = f_{xxx}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) \right) = \frac{\partial^3 f}{\partial y \partial x \partial x} = f_{xxy}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \right) = \frac{\partial^3 f}{\partial x \partial y \partial x} = f_{xyx}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \right) = \frac{\partial^3 f}{\partial y \partial y \partial x} = f_{xyy}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) \right) = \frac{\partial^3 f}{\partial x \partial x \partial y} = f_{yxx}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) \right) = \frac{\partial^3 f}{\partial y \partial x \partial y} = f_{yxy}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) \right) = \frac{\partial^3 f}{\partial x \partial y \partial y} = f_{yyx}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) \right) = \frac{\partial^3 f}{\partial y^3} = f_{yyy}$$

Example of Partial Derivatives

Example 10: If $u = e^{3xyz}$ show that $\frac{\partial^3 u}{\partial x \partial y \partial z} = (3 + 27xyz + 27x^2y^2z^2)e^{3xyz}$.

Solution: Here, $u = e^{3xyz}$

$$\begin{aligned}\Rightarrow \frac{\partial u}{\partial z} &= \frac{\partial}{\partial z}(u) \\ &= \frac{\partial}{\partial z}(e^{3xyz}) \\ &= e^{3xyz} \cdot \frac{\partial}{\partial z}(3xy z) \\ &= e^{3xyz} \cdot 3xy \cdot 1 \\ \Rightarrow \frac{\partial u}{\partial z} &= 3xy \cdot e^{3xyz}\end{aligned}$$

$$\begin{aligned}\Rightarrow \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial z} \right) &= \frac{\partial}{\partial y} (3x y \cdot e^{3xyz}) \\ &= 3x \frac{\partial}{\partial y} (y \cdot e^{3xyz}) \\ &= 3x \{ y \cdot e^{3xyz} \cdot 3xz + e^{3xyz} \cdot 1 \} \\ &= 3x \{ 3xyz \cdot e^{3xyz} + e^{3xyz} \} \\ &= \{ 9x^2yz \cdot e^{3xyz} + 3x \cdot e^{3xyz} \}\end{aligned}$$

Example of Partial Derivatives

Example 10: If $u = e^{3xyz}$ show that $\frac{\partial^3 u}{\partial x \partial y \partial z} = (3 + 27xyz + 27x^2y^2z^2)e^{3xyz}$.

Solution Continue:

$$\Rightarrow \frac{\partial^2 u}{\partial y \partial z} = e^{3xyz} \cdot (9x^2yz + 3x)$$

$$\begin{aligned}\Rightarrow \frac{\partial^3 u}{\partial x \partial y \partial z} &= \frac{\partial}{\partial x} \left(\frac{\partial^2 u}{\partial y \partial z} \right) = \{e^{3xyz} \cdot (18xyz + 3) + (9x^2yz + 3x) \cdot e^{3xyz} \cdot 3yz\} \\ &= \{18xyz + 3 + (9x^2yz + 3x) \cdot 3yz\} e^{3xyz} \\ &= (18xyz + 3 + 27x^2y^2z^2 + 9xyz) e^{3xyz}\end{aligned}$$

$$\Rightarrow \frac{\partial^3 u}{\partial x \partial y \partial z} = (3 + 27xyz + 27x^2y^2z^2) e^{3xyz}$$

Example of Partial Derivatives

Example 12: If $u = \log(\tan x + \tan y + \tan z)$ then prove that

$$\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2.$$

Solution: Here, $u = \log(\tan x + \tan y + \tan z)$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{1}{\tan x + \tan y + \tan z} \cdot \frac{\partial}{\partial x} (\tan x + \tan y + \tan z)$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{1}{\tan x + \tan y + \tan z} \cdot \sec^2 x$$

Similarly we get

$$\Rightarrow \frac{\partial u}{\partial y} = \frac{\sec^2 y}{\tan x + \tan y + \tan z} \quad \& \quad \Rightarrow \frac{\partial u}{\partial z} = \frac{\sec^2 z}{\tan x + \tan y + \tan z}$$

Example of Partial Derivatives

Example 12: If $u = \log(\tan x + \tan y + \tan z)$ then prove that

$$\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2.$$

Solution Continue:

$$\text{L.H.S.} = \sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z}$$

$$\begin{aligned} &= \frac{\sin 2x \cdot \sec^2 x}{\tan x + \tan y + \tan z} + \frac{\sin 2y \cdot \sec^2 y}{\tan x + \tan y + \tan z} + \frac{\sin 2z \cdot \sec^2 z}{\tan x + \tan y + \tan z} \\ &= \frac{\sin 2x \cdot \sec^2 x}{\tan x + \tan y + \tan z} + \frac{2 \tan y}{\tan x + \tan y + \tan z} + \frac{2 \tan z}{\tan x + \tan y + \tan z} \end{aligned}$$

$$\begin{aligned} \text{Now, } \sin 2x \cdot \sec^2 x &= 2 \sin x \cancel{\cos x} \cdot \frac{1}{\cancel{\cos^2 x}} \\ &= 2 \frac{\sin x}{\cos x} = 2 \tan x \end{aligned}$$

Example of Partial Derivatives

Example 12: If $u = \log(\tan x + \tan y + \tan z)$ then prove that

$$\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2.$$

Solution Continue:

$$= \frac{2}{\cancel{\tan x + \tan y + \tan z}} \cancel{(\tan x + \tan y + \tan z)}$$

$$= 2 = \text{R.H.S.}$$

$$\Rightarrow \sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$$

Example of Partial Derivatives

Example 19: If $u = \log(x^3 + y^3 - x^2y - xy^2)$, prove that

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right)^2 u = -\frac{4}{(x+y)^2}.$$

Solution: Here, $u = \log(x^3 - x^2y - xy^2 + y^3)$
 $= \log(x^2(x - y) - y^2(x - y))$
 $= \log((x - y)(x^2 - y^2))$
 $= \log((x - y)(x + y)(x - y))$
 $= \log((x + y)(x - y)^2)$
 $= \log(x + y) + \log(x - y)^2$
 $\Rightarrow u = \log(x + y) + 2 \log(x - y)$

Example of Partial Derivatives

Example 19: If $u = \log(x^3 + y^3 - x^2y - xy^2)$, prove that

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right)^2 u = -\frac{4}{(x+y)^2}.$$

Solution Continue: Here, $u = \log(x+y) + 2\log(x-y)$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{1}{x+y} + \frac{2}{x-y} \quad \& \quad \frac{\partial u}{\partial y} = \frac{1}{x+y} - \frac{2}{x-y}$$

$$\Rightarrow \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{1}{x+y} + \cancel{\frac{2}{x-y}} + \frac{1}{x+y} - \cancel{\frac{2}{x-y}}$$

$$\Rightarrow \boxed{\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{2}{x+y}} \quad \text{-----} \rightarrow (1)$$

Example of Partial Derivatives

Example 19: If $u = \log(x^3 + y^3 - x^2y - xy^2)$, prove that

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right)^2 u = -\frac{4}{(x+y)^2}.$$

Solution Continue:

$$\begin{aligned} \text{Now, } \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right)^2 u &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) u \\ &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \right) \\ &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right) \left(\frac{2}{x+y} \right) \quad \because \text{From (1)} \end{aligned}$$

Example of Partial Derivatives

Example 19: If $u = \log(x^3 + y^3 - x^2y - xy^2)$, prove that

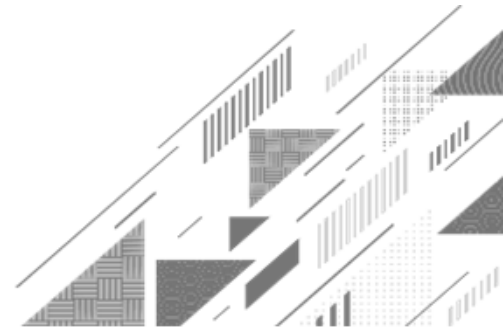
$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right)^2 u = -\frac{4}{(x+y)^2}.$$

Solution Continue:

$$\begin{aligned} &= \frac{\partial}{\partial x} \left(\frac{2}{x+y} \right) + \frac{\partial}{\partial y} \left(\frac{2}{x+y} \right) \\ &= -\frac{2}{(x+y)^2} - \frac{2}{(x+y)^2} \end{aligned}$$

$$\Rightarrow \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} \right)^2 u = -\frac{4}{(x+y)^2}$$

Method:4 ➡ Chain Rule

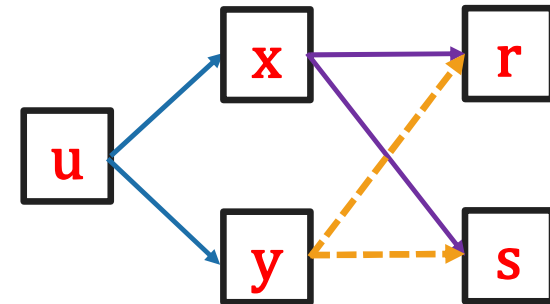


Chain Rule

► Case 1: If $u = f(x, y)$ and $x = g(r, s)$ & $y = h(r, s)$, then

$$\rightarrow \frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$\rightarrow \frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s}$$



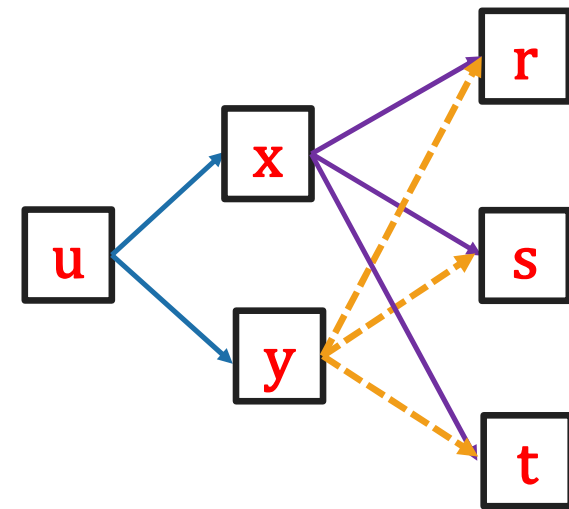
Chain Rule

► Case 2: If $u = f(x, y)$ and $x = g(r, s, t)$ & $y = h(r, s, t)$, then

$$\rightarrow \frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r}$$

$$\rightarrow \frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s}$$

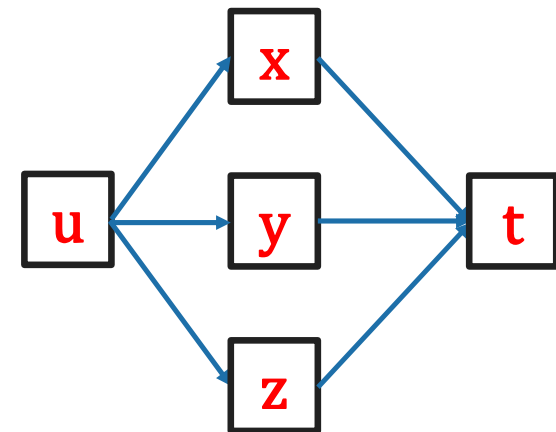
$$\rightarrow \frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial t}$$



Chain Rule

► Case 3: If $u = f(x, y, z)$ and $x = g(t)$, $y = h(t)$ & $z = k(t)$, then

$$\rightarrow \frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt}$$

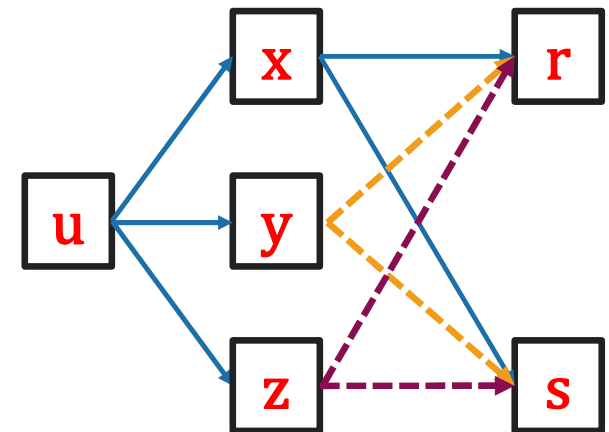


Chain Rule

► Case 4: If $u = f(x, y, z)$ and $x = g(r, s)$, $y = h(r, s)$ & $z = k(r, s)$, then

$$\rightarrow \frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial r}$$

$$\rightarrow \frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial s}$$



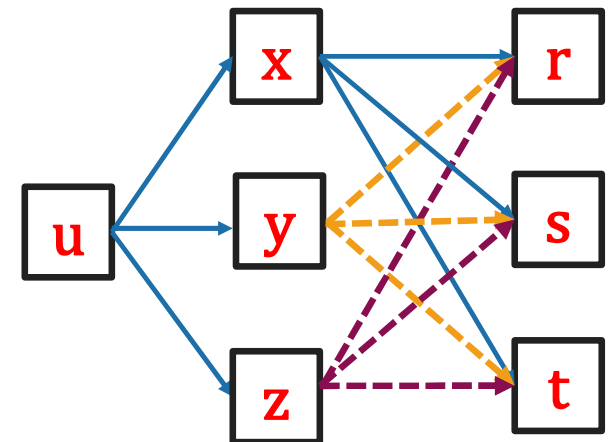
Chain Rule

► Case 5: If $u = f(x, y, z)$ and $x = g(r, s, t)$, $y = h(r, s, t)$ & $z = k(r, s, t)$, then

$$\rightarrow \frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial r} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial r}$$

$$\rightarrow \frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial s}$$

$$\rightarrow \frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial t} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial t}$$



Example of Chain Rule

Example 1: Find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ in terms of r and s , if $w = x + 2y + z^2$.

Where, $x = \frac{r}{s}$, $y = r^2 + \ln s$ & $z = 2r$.

Solution: Here, $w = x + 2y + z^2$.

$$\Rightarrow \frac{\partial w}{\partial x} = 1, \quad \frac{\partial w}{\partial y} = 2 \quad \& \quad \frac{\partial w}{\partial z} = 2z$$

$$\rightarrow x = \frac{r}{s}$$

$$\Rightarrow \frac{\partial x}{\partial r} = \frac{1}{s}$$

$$\& \quad \frac{\partial x}{\partial s} = \frac{-r}{s^2}$$

$$\rightarrow y = r^2 + \ln s$$

$$\Rightarrow \frac{\partial y}{\partial r} = 2r$$

$$\& \quad \frac{\partial y}{\partial s} = \frac{1}{s}$$

$$\rightarrow z = 2r$$

$$\Rightarrow \frac{\partial z}{\partial r} = 2$$

$$\& \quad \frac{\partial z}{\partial s} = 0$$

Example of Chain Rule

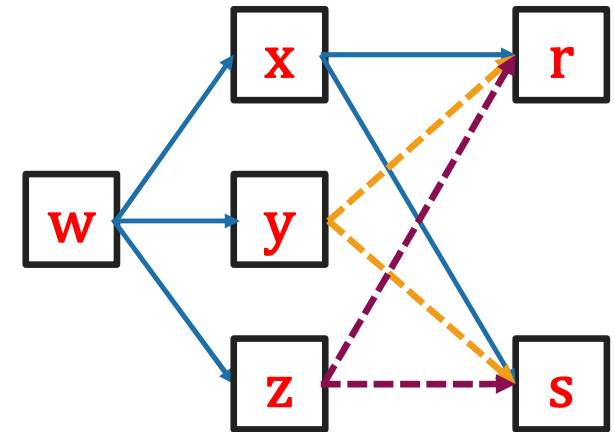
Example 1: Find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ in terms of r and s , if $w = x + 2y + z^2$.

Where, $x = \frac{r}{s}$, $y = r^2 + \ln s$ & $z = 2r$.

Solution Continue:

$$\begin{aligned} \Rightarrow \frac{\partial w}{\partial r} &= \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial r} \\ &= (1) \left(\frac{1}{s} \right) + (2)(2r) + (2z)(2) \\ &= \frac{1}{s} + 4r + \{2(2r)\}(2) = \frac{1}{s} + 4r + 8r \end{aligned}$$

$$\Rightarrow \frac{\partial w}{\partial r} = \frac{1}{s} + 12r$$



Example of Chain Rule

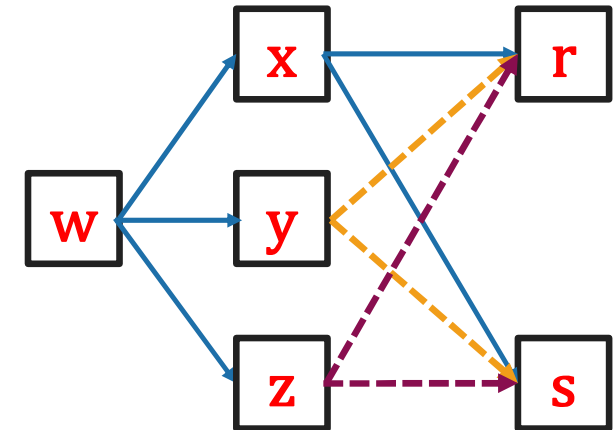
Example 1: Find $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ in terms of r and s , if $w = x + 2y + z^2$.

Where, $x = \frac{r}{s}$, $y = r^2 + \ln s$ & $z = 2r$.

Solution Continue:

$$\begin{aligned} \Rightarrow \frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial s} \\ &= (1) \left(\frac{-r}{s^2} \right) + (2) \left(\frac{1}{s} \right) + (2z)(0) \\ &= \frac{-r}{s^2} + \frac{2}{s} + 0 \end{aligned}$$

$$\Rightarrow \frac{\partial w}{\partial s} = \frac{2}{s} - \frac{r}{s^2}$$



Example of Chain Rule

Example 3: If $z = x^2y + 3xy^4$, where $x = \sin 2t$ & $y = \cos t$, find $\frac{dz}{dt}$ when $t = 0$.

Solution:

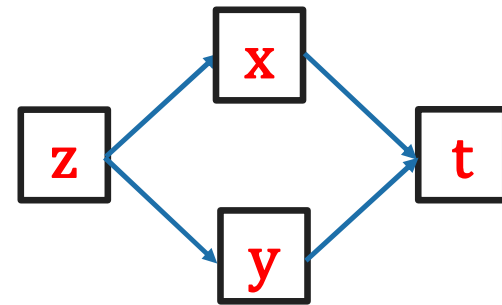
$$\rightarrow \frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

$$= (2xy + 3y^4) \cdot 2 \cos 2t + (x^2 + 12xy^3) \cdot (-\sin t)$$

$$\text{When } t = 0 \Rightarrow x = \sin 0 = 0 \text{ \& } y = \cos 0 = 1$$

$$\Rightarrow \left\{ \frac{\partial z}{\partial t} \right\}_{\substack{t=0 \\ x=0, y=1}} = \{2(0)(1) + 3(1)^4\} \cdot 2 \cos 0 - \{2(0)(1) + 3(1)^4\} \cdot (-\sin 0)$$
$$= 3(2)$$

$$\Rightarrow \left\{ \frac{\partial z}{\partial t} \right\}_{t=0} = 6$$



Example of Chain Rule

Example 6: If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ then show that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.

Solution: Let, $u = f(r, s) = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$

$$\begin{aligned}\Rightarrow r &= \frac{y-x}{xy} \\ &= \frac{\cancel{y}}{x\cancel{y}} - \frac{\cancel{x}}{\cancel{x}y}\end{aligned}$$

$$\Rightarrow r = \frac{1}{x} - \frac{1}{y}$$

$$\begin{aligned}\Rightarrow s &= \frac{z-x}{xz} \\ &= \frac{\cancel{z}}{x\cancel{z}} - \frac{\cancel{x}}{\cancel{x}z}\end{aligned}$$

$$\Rightarrow s = \frac{1}{x} - \frac{1}{z}$$

Example of Chain Rule

Example 6: If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ then show that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.

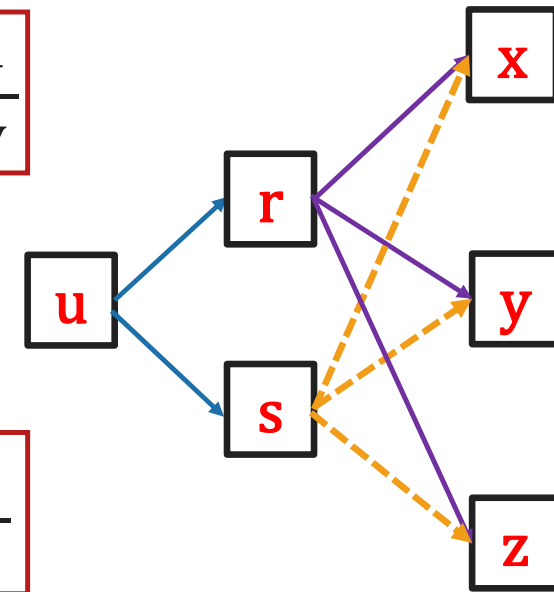
Solution Continue:

$$\begin{aligned} \Rightarrow \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} \\ &= \frac{\partial u}{\partial r} \cdot \left(-\frac{1}{x^2}\right) + \frac{\partial u}{\partial s} \cdot \left(-\frac{1}{x^2}\right) \end{aligned}$$

$$\Rightarrow \frac{\partial u}{\partial x} = -\frac{1}{x^2} \cdot \frac{\partial u}{\partial r} - \frac{1}{x^2} \cdot \frac{\partial u}{\partial s}$$

$$r = \frac{1}{x} - \frac{1}{y}$$

$$s = \frac{1}{x} - \frac{1}{z}$$



Example of Chain Rule

Example 6: If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ then show that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.

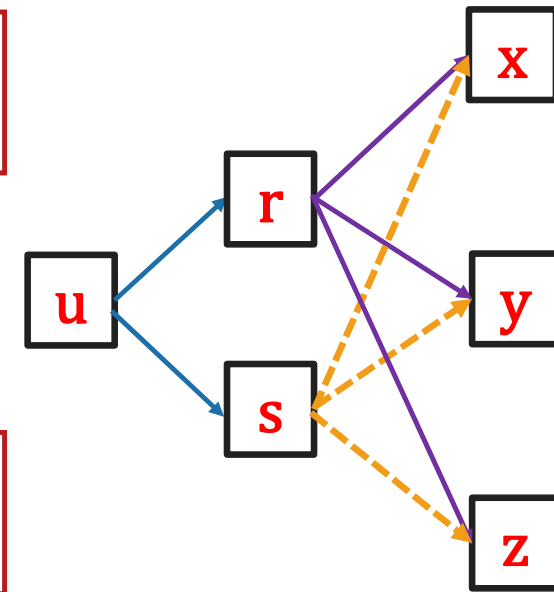
Solution Continue:

$$\begin{aligned} \Rightarrow \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} \\ &= \frac{\partial u}{\partial r} \cdot \left(\frac{1}{y^2}\right) + \frac{\partial u}{\partial s} \cdot (0) \end{aligned}$$

$$\Rightarrow \boxed{\frac{\partial u}{\partial y} = \frac{1}{y^2} \cdot \frac{\partial u}{\partial r}}$$

$$\boxed{r = \frac{1}{x} - \frac{1}{y}}$$

$$\boxed{s = \frac{1}{x} - \frac{1}{z}}$$



Example of Chain Rule

Example 6: If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ then show that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.

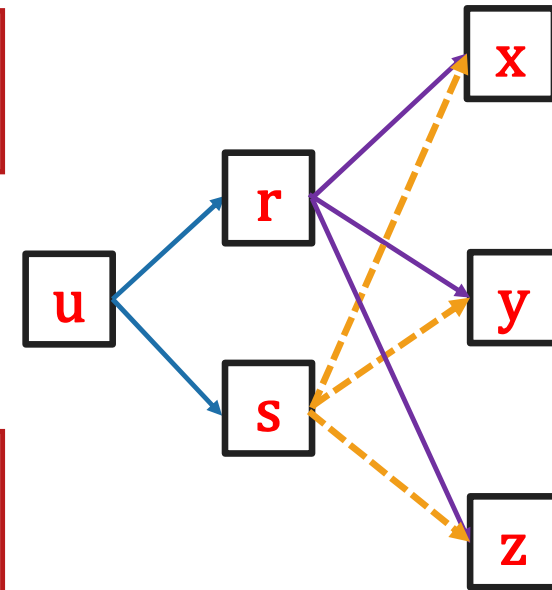
Solution Continue:

$$\begin{aligned} \Rightarrow \frac{\partial u}{\partial z} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial z} \\ &= \frac{\partial u}{\partial r} \cdot (0) + \frac{\partial u}{\partial s} \cdot \left(\frac{1}{z^2}\right) \end{aligned}$$

$$\Rightarrow \frac{\partial u}{\partial z} = \frac{1}{z^2} \cdot \frac{\partial u}{\partial s}$$

$$r = \frac{1}{x} - \frac{1}{y}$$

$$s = \frac{1}{x} - \frac{1}{z}$$



Example of Chain Rule

Example 6: If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ then show that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$.

Solution Continue:

$$\begin{aligned} &\rightarrow x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} \\ &= \cancel{x^2} \left(-\frac{1}{\cancel{x^2}} \cdot \frac{\partial u}{\partial r} - \frac{1}{\cancel{x^2}} \cdot \frac{\partial u}{\partial s} \right) + \cancel{y^2} \left(\frac{1}{\cancel{y^2}} \cdot \frac{\partial u}{\partial r} \right) + \cancel{z^2} \left(\frac{1}{\cancel{z^2}} \cdot \frac{\partial u}{\partial s} \right) \\ &= -\frac{\cancel{\partial u}}{\cancel{\partial r}} - \frac{\cancel{\partial u}}{\cancel{\partial s}} + \frac{\cancel{\partial u}}{\cancel{\partial r}} + \frac{\cancel{\partial u}}{\cancel{\partial s}} = 0 \\ &\Rightarrow \boxed{x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0} \end{aligned}$$

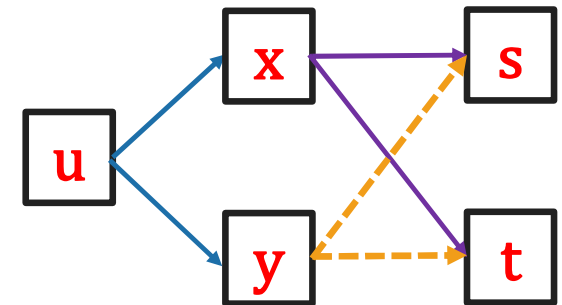
Example of Chain Rule

Example 10: If $u = f(x, y)$, where $x = e^s \cos t$ and $y = e^s \sin t$, show that

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = e^{-2s} \left[\left(\frac{\partial u}{\partial s}\right)^2 + \left(\frac{\partial u}{\partial t}\right)^2 \right].$$

Solution:

$$\begin{aligned} \Rightarrow \frac{\partial u}{\partial s} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s} \\ &= \frac{\partial u}{\partial x} \cdot e^s \cos t + \frac{\partial u}{\partial y} \cdot e^s \sin t \\ \Rightarrow \frac{\partial u}{\partial s} &= e^s \left(\frac{\partial u}{\partial x} \cdot \cos t + \frac{\partial u}{\partial y} \cdot \sin t \right) \end{aligned}$$



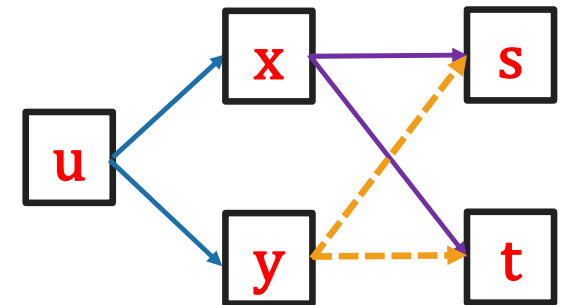
Example of Chain Rule

Example 10: If $u = f(x, y)$, where $x = e^s \cos t$ and $y = e^s \sin t$, show that

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = e^{-2s} \left[\left(\frac{\partial u}{\partial s}\right)^2 + \left(\frac{\partial u}{\partial t}\right)^2 \right].$$

Solution Continue:

$$\begin{aligned} \Rightarrow \frac{\partial u}{\partial t} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial t} \\ &= \frac{\partial u}{\partial x} \cdot (-e^s \sin t) + \frac{\partial u}{\partial y} \cdot e^s \cos t \\ \Rightarrow \frac{\partial u}{\partial t} &= e^s \left(\frac{\partial u}{\partial y} \cdot \cos t - \frac{\partial u}{\partial x} \cdot \sin t \right) \end{aligned}$$



Example of Chain Rule

Solution Continue:

$$\text{R. H. S.} = e^{-2s} \left[\left(\frac{\partial u}{\partial s} \right)^2 + \left(\frac{\partial u}{\partial t} \right)^2 \right]$$

$$\begin{aligned} &= e^{-2s} \left[\left\{ e^s \left(\frac{\partial u}{\partial x} \cdot \cos t + \frac{\partial u}{\partial y} \cdot \sin t \right) \right\}^2 + \left\{ e^s \left(\frac{\partial u}{\partial y} \cdot \cos t - \frac{\partial u}{\partial x} \cdot \sin t \right) \right\}^2 \right] \\ &= e^{-2s} \left[e^{2s} \left(\frac{\partial u}{\partial x} \cdot \cos t + \frac{\partial u}{\partial y} \cdot \sin t \right)^2 + e^{2s} \left(\frac{\partial u}{\partial y} \cdot \cos t - \frac{\partial u}{\partial x} \cdot \sin t \right)^2 \right] \\ &= \cancel{e^{-2s}} \cdot \cancel{e^{2s}} \left[\left(\frac{\partial u}{\partial x} \right)^2 \cdot \cos^2 t + \cancel{2 \cos t \sin t \frac{\partial u}{\partial x} \frac{\partial u}{\partial y}} + \left(\frac{\partial u}{\partial y} \right)^2 \cdot \sin^2 t \right. \\ &\quad \left. + \left(\frac{\partial u}{\partial y} \right)^2 \cdot \cos^2 t - \cancel{2 \cos t \sin t \frac{\partial u}{\partial x} \frac{\partial u}{\partial y}} + \left(\frac{\partial u}{\partial x} \right)^2 \cdot \sin^2 t \right] \end{aligned}$$

Example of Chain Rule

Solution Continue:

$$= \left(\frac{\partial u}{\partial x} \right)^2 \cdot \cos^2 t + \left(\frac{\partial u}{\partial y} \right)^2 \cdot \sin^2 t + \left(\frac{\partial u}{\partial y} \right)^2 \cdot \cos^2 t + \left(\frac{\partial u}{\partial x} \right)^2 \cdot \sin^2 t$$

$$= \left(\frac{\partial u}{\partial x} \right)^2 \cdot (\cos^2 t + \sin^2 t) + \left(\frac{\partial u}{\partial y} \right)^2 \cdot (\sin^2 t + \cos^2 t)$$

$$= \left(\frac{\partial u}{\partial x} \right)^2 \cdot (1) + \left(\frac{\partial u}{\partial y} \right)^2 \cdot (1) = \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = \text{L. H. S.}$$

$$\Rightarrow \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right] = e^{-2s} \left[\left(\frac{\partial u}{\partial s} \right)^2 + \left(\frac{\partial u}{\partial t} \right)^2 \right]$$

Example of Chain Rule

Example 11: If $u = f(r)$ and $r^2 = x^2 + y^2 + z^2$ then show that

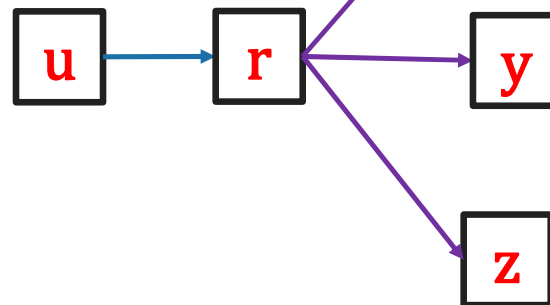
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f''(r) + \frac{2}{r} f'(r).$$

Solution:

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{du}{dr} \cdot \frac{\partial r}{\partial x}$$

= .

$$\Rightarrow \boxed{\frac{\partial u}{\partial x} = f'(r) \cdot x \cdot \frac{1}{r}}$$



$$\Rightarrow u = f(r)$$

$$\Rightarrow \frac{du}{dr} = f'(r)$$

$$\Rightarrow r^2 = x^2 + y^2 + z^2$$

$$\Rightarrow \cancel{r} \frac{\partial r}{\partial x} = \cancel{r} x + 0 + 0$$

$$\Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$$

Example of Chain Rule

Solution Continue:

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

$$\rightarrow \frac{\partial u}{\partial x} = f'(r) \cdot x \cdot \frac{1}{r}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = f'(r) \cdot x \cdot \left(-\frac{1}{r^2} \cdot \frac{\partial r}{\partial x} \right) + f'(r) \cdot 1 \cdot \frac{1}{r} + \left(f''(r) \cdot \frac{\partial r}{\partial x} \right) \cdot x \cdot \frac{1}{r}$$

$$= f'(r) \cdot x \cdot \left(-\frac{1}{r^2} \cdot \frac{x}{r} \right) + f'(r) \cdot 1 \cdot \frac{1}{r} + \left(f''(r) \cdot \frac{x}{r} \right) \cdot x \cdot \frac{1}{r}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = -\frac{x^2 \cdot f'(r)}{r^3} + \frac{f'(r)}{r} + \frac{x^2 \cdot f''(r)}{r^2}$$

Example of Chain Rule

Solution Continue:

$$\frac{\partial^2 u}{\partial x^2} = -\frac{x^2 \cdot f'(r)}{r^3} + \frac{f'(r)}{r} + \frac{x^2 \cdot f''(r)}{r^2}$$

Similarly we get,

$$\frac{\partial^2 u}{\partial y^2} = -\frac{y^2 \cdot f'(r)}{r^3} + \frac{f'(r)}{r} + \frac{y^2 \cdot f''(r)}{r^2}$$

and

$$\frac{\partial^2 u}{\partial z^2} = -\frac{z^2 \cdot f'(r)}{r^3} + \frac{f'(r)}{r} + \frac{z^2 \cdot f''(r)}{r^2}$$

$$r^2 = x^2 + y^2 + z^2$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$

$$= -\frac{f'(r)}{r^3} (x^2 + y^2 + z^2) + 3 \frac{f'(r)}{r} + \frac{f''(r)}{r^2} (x^2 + y^2 + z^2)$$

$$= -\frac{f'(r)}{r^3} (r^2) + \frac{3 f'(r)}{r} + \frac{f''(r)}{r^2} (r^2)$$

Example of Chain Rule

Example 11: If $u = f(r)$ and $r^2 = x^2 + y^2 + z^2$ then show that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f''(r) + \frac{2}{r} f'(r).$$

Solution Continue:

$$\begin{aligned} \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} &= -\frac{f'(r)}{r^3} (\cancel{r^2}) + \frac{3 f'(r)}{r} + \frac{f''(r)}{\cancel{r^2}} (\cancel{r^2}) \\ &= -\frac{f'(r)}{r} + \frac{3 f'(r)}{r} + f''(r) \end{aligned}$$

$$\Rightarrow \boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f''(r) + \frac{2}{r} f'(r)}$$

Total Differentiation

→ If $u = f(x, y)$ then total differentiation of u is given by following formula.

$$du = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

Here du , dx and dy are called total(exact) differentials of u , x and y respectively.

→ If $u = f(x, y, z)$ then total differentiation of u is given by following formula.

$$du = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

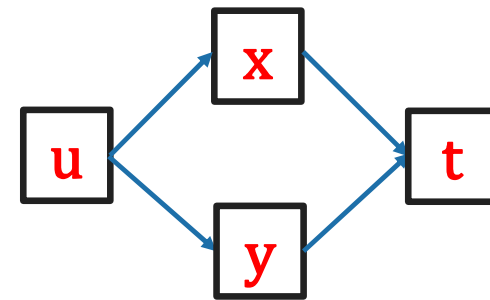
→ If $u = f(x_1, x_2, \dots, x_n)$ then total differentiation of u is given by following formula.

$$du = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_n} dx_n$$

Example of Total Differentiation

Example 13: Find $\frac{du}{dt}$ as total derivative for $u = 4x + xy - y^2$, $x = \cos 3t$ & $y = \sin 3t$.

Solution: Here, $u = 4x + xy - y^2$, $x = \cos 3t$ & $y = \sin 3t$.



$$\rightarrow \frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

$$= (4 + y) \cdot (-3 \sin 3t) + (x - 2y) \cdot 3 \cos 3t$$

$$= (4 + \sin 3t) \cdot (-3 \sin 3t) + (\cos 3t - 2 \sin 3t) \cdot 3 \cos 3t$$

$$= -12 \sin 3t - \underline{3 \sin^2 3t} + 3 \cos^2 3t - 6 \sin 3t \cos 3t$$

$$= -12 \sin 3t + 3(\cos^2 3t - \sin^2 3t) - 3 \cdot 2 \sin 3t \cos 3t$$

Example of Total Differentiation

Example 13: Find $\frac{du}{dt}$ as total derivative for $u = 4x + xy - y^2$, $x = \cos 3t$ & $y = \sin 3t$.

Solution Continue:

$$\Rightarrow \frac{du}{dt} = -12 \sin 3t + 3(\cos^2 3t - \sin^2 3t) - 3 \cdot 2 \sin 3t \cos 3t$$

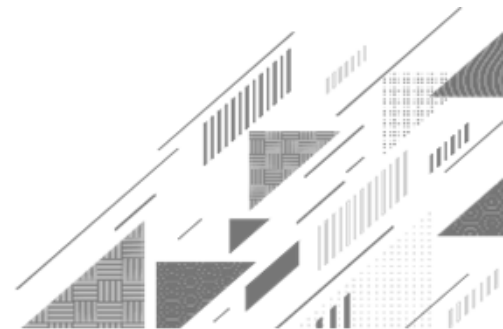
$$= -12 \sin 3t + 3(\cos 6t) - 3(\sin 6t)$$

$$\Rightarrow \frac{du}{dt} = -12 \sin 3t + 3 \cos 6t - 3 \sin 6t$$

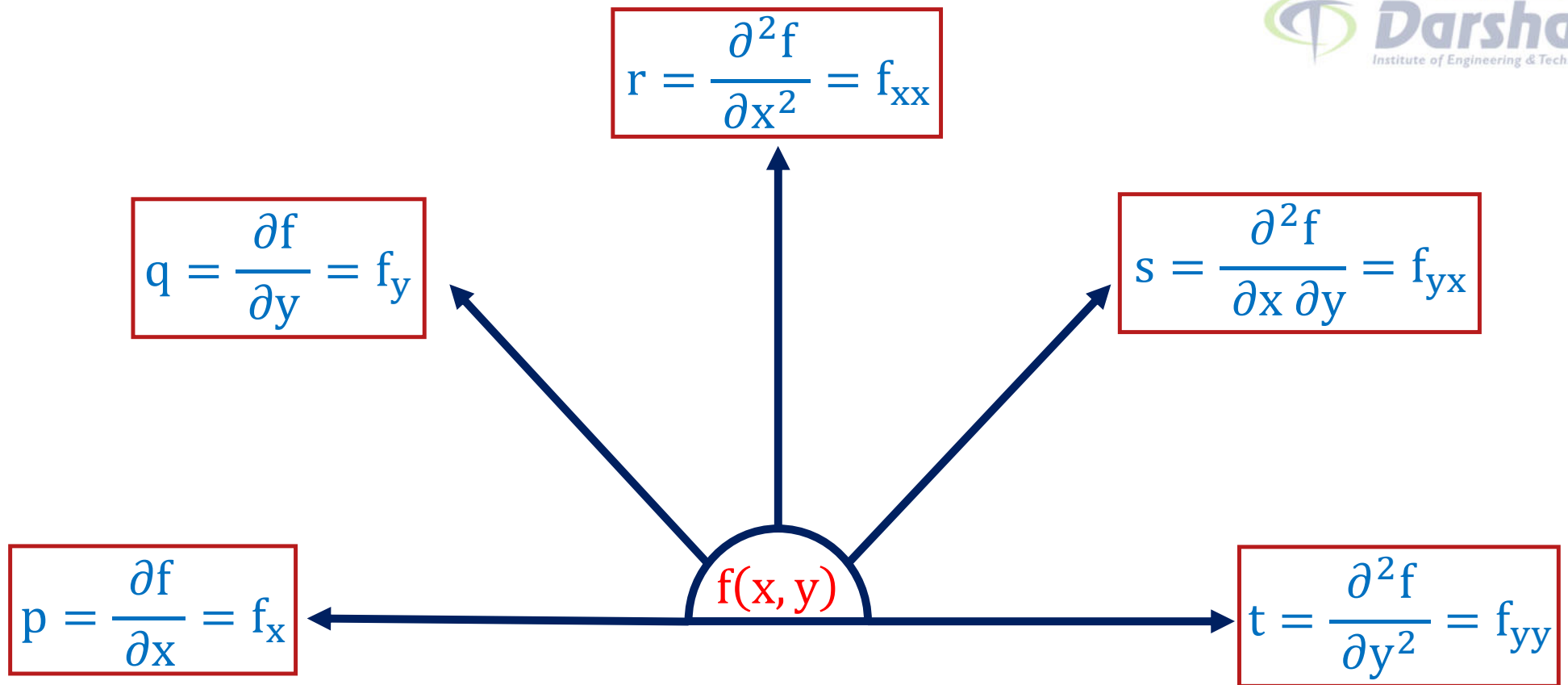
$$\cos^2 x - \sin^2 x = \cos 2x$$

$$2 \sin x \cos x = \sin 2x$$

Method:5 ➡ Implicit Function



Important Notations



Implicit Function

→ A function in which **one variable** can **not** be **express** in terms of other variables is called an **implicit function**.

For Example $x^3 + y^3 = 6xy$

→ First order differential coefficient of an implicit function:

Let $f(x, y) = c$ be an implicit function then

$$\boxed{\frac{dy}{dx} = -\frac{p}{q}}, \text{ where } q \neq 0, p = \frac{\partial f}{\partial x} \text{ \& } q = \frac{\partial f}{\partial y}.$$

Example of an Implicit Function

Example 1: By using partial derivatives find the value of $\frac{dy}{dx}$ at $(0, \log 2)$ for $f(x, y) = xe^y + \sin(xy) + y - \log 2$.

Solution: Here, $f(x, y) = xe^y + \sin(xy) + y - \log 2$.

$$\Rightarrow p = \frac{\partial f}{\partial x} = e^y + y \cos xy$$

$$\& q = \frac{\partial f}{\partial y} = x e^y + x \cos xy + 1$$

$$\Rightarrow \frac{dy}{dx} = -\frac{p}{q} = -\frac{e^y + y \cos xy}{x e^y + x \cos xy + 1}$$

$$\Rightarrow \frac{dy}{dx}_{(0, \log 2)} = -\frac{(e^{\log_e 2} + \log 2 \cos 0)}{0 + 0 + 1}$$

$$= -(2 + \log 2)$$

$$\Rightarrow \frac{dy}{dx}_{(0, \log 2)} = -(2 + \log 2)$$

Implicit Function

→ Second order differential coefficient of an implicit function:

Let $f(x, y) = c$ be an implicit function then

$$\frac{d^2y}{dx^2} = - \left[\frac{q^2r - 2pqs + p^2t}{q^3} \right]$$

OR

$$\frac{d^2y}{dx^2} = \frac{\begin{vmatrix} r & s & p \\ s & t & q \\ p & q & 0 \end{vmatrix}}{q^3}$$

→ $q \neq 0, p = \frac{\partial f}{\partial x}, q = \frac{\partial f}{\partial y}, r = \frac{\partial^2 f}{\partial x^2}, s = \frac{\partial^2 f}{\partial x \partial y} \text{ \& } t = \frac{\partial^2 f}{\partial y^2}$

Example of an Implicit Function

Example 6: If $x^3 + y^3 = 6xy$, then find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

Solution:

Here, $x^3 + y^3 = 6xy$.

$$\Rightarrow x^3 + y^3 - 6xy = 0$$

$$\Rightarrow f(x, y) = x^3 + y^3 - 6xy$$

$$\rightarrow p = \frac{\partial f}{\partial x} = 3x^2 - 6y$$

$$\rightarrow q = \frac{\partial f}{\partial y} = 3y^2 - 6x$$

$$\rightarrow r = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (3x^2 - 6y) = 6x$$

$$\rightarrow s = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (3y^2 - 6x) = -6$$

$$\rightarrow t = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (3y^2 - 6x) = 6y$$

Example of an Implicit Function

Example 6: If $x^3 + y^3 = 6xy$, then find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

Solution Continue: $p = 3x^2 - 6y$, $q = 3y^2 - 6x$, $r = 6x$, $s = -6$ & $t = 6y$

$$\rightarrow \frac{dy}{dx} = -\frac{p}{q} = -\frac{3x^2 - 6y}{3y^2 - 6x} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{\cancel{3}(2y - x^2)}{\cancel{3}(y^2 - 2x)} \Rightarrow \boxed{\frac{dy}{dx} = \frac{2y - x^2}{y^2 - 2x}}$$

$$\begin{aligned} \rightarrow \frac{d^2y}{dx^2} &= -\left[\frac{q^2r - 2pqs + p^2t}{q^3} \right] \\ &= -\left[\frac{(3y^2 - 6x)^2(6x) - 2(3x^2 - 6y)(3y^2 - 6x)(-6) + (3x^2 - 6y)^2(6y)}{(3y^2 - 6x)^3} \right] \end{aligned}$$

Example of an Implicit Function

Solution Continue:

$$\begin{aligned}
 &= - \left[\frac{(3y^2 - 6x)^2(6x) + 12(3x^2 - 6y)(3y^2 - 6x) + (3x^2 - 6y)^2(6y)}{(3y^2 - 6x)^3} \right] \\
 &= - \left[\frac{9(y^2 - 2x)^2(6x) + 2 \cdot 6(9)(x^2 - 2y)(y^2 - 2x) + 9(x^2 - 2y)^2(6y)}{27(y^2 - 2x)^3} \right] \\
 &= - \frac{\cancel{54}^2}{\cancel{27}} \left[\frac{(y^4 - 4xy^2 + 4x^2)x + 2(x^2y^2 - 2x^3 - 2y^3 + 4xy) + (x^4 - 4x^2y + 4y^2)y}{(y^2 - 2x)^3} \right] \\
 &= -2 \left[\frac{xy^4 - 4x^2y^2 + \cancel{4x^3} + 2x^2y^2 - \cancel{4x^3} - \cancel{4y^3} + 8xy + x^4y - 4x^2y^2 + \cancel{4y^3}}{(y^2 - 2x)^3} \right]
 \end{aligned}$$

Example of an Implicit Function

Example 6: If $x^3 + y^3 = 6xy$, then find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

Solution Continue:

$$\begin{aligned} &= -2 \left[\frac{xy^4 - 4x^2y^2 + 2x^2y^2 + 8xy + x^4y - 4x^2y^2}{(y^2 - 2x)^3} \right] \\ &= -2 \left[\frac{xy(x^3 + y^3) - 6x^2y^2 + 8xy}{(y^2 - 2x)^3} \right] = -2 \left[\frac{xy(\quad) - 6x^2y^2 + 8xy}{(y^2 - 2x)^3} \right] \\ &= -2 \left[\frac{\cancel{6x^2y^2} - \cancel{6x^2y^2} + 8xy}{(y^2 - 2x)^3} \right] = -2 \left[\frac{8xy}{(y^2 - 2x)^3} \right] \end{aligned}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \left[\frac{-16xy}{(y^2 - 2x)^3} \right]$$

Partial Differentiation of an Implicit Function

→ If $f(x, y, z) = 0$ is an implicit function then

$$\frac{\partial z}{\partial x} = -\frac{\partial f / \partial x}{\partial f / \partial z} = -\frac{f_x}{f_z}$$

$$\frac{\partial z}{\partial y} = -\frac{\partial f / \partial y}{\partial f / \partial z} = -\frac{f_y}{f_z}$$

$$\frac{\partial x}{\partial y} = -\frac{\partial f / \partial y}{\partial f / \partial x} = -\frac{f_y}{f_x}$$

$$\frac{\partial x}{\partial z} = -\frac{\partial f / \partial z}{\partial f / \partial x} = -\frac{f_z}{f_x}$$

$$\frac{\partial y}{\partial z} = -\frac{\partial f / \partial z}{\partial f / \partial y} = -\frac{f_z}{f_y}$$

$$\frac{\partial y}{\partial x} = -\frac{\partial f / \partial x}{\partial f / \partial y} = -\frac{f_x}{f_y}$$

Example of an Implicit Function

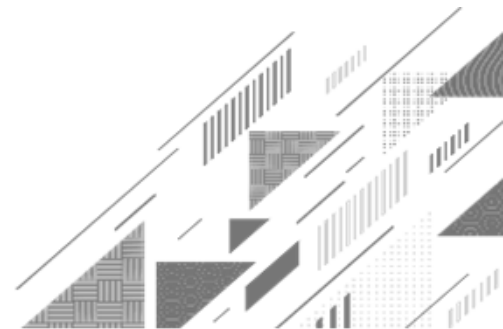
Example 9: Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if z defined implicitly as a function of x and y by the equation $x^3 + y^3 + z^3 + 6xyz = 1$.

Solution: Here, $x^3 + y^3 + z^3 + 6xyz = 1$.

$$\Rightarrow f(x, y, z) = x^3 + y^3 + z^3 + 6xyz - 1$$

$$\begin{array}{l} \rightarrow \frac{\partial f}{\partial x} = 3x^2 + 6yz \\ \rightarrow \frac{\partial f}{\partial y} = 3y^2 + 6xz \\ \rightarrow \frac{\partial f}{\partial z} = 3z^2 + 6xy \end{array} \quad \left| \quad \begin{array}{l} \frac{\partial z}{\partial x} = -\frac{\partial f / \partial x}{\partial f / \partial z} = -\left[\frac{\quad}{\quad} \right] = -\left[\frac{x^2 + 2yz}{z^2 + 2xy} \right] \\ \frac{\partial z}{\partial y} = -\frac{\partial f / \partial y}{\partial f / \partial z} = -\left[\frac{\quad}{\quad} \right] = -\left[\frac{y^2 + 2xz}{z^2 + 2xy} \right] \end{array} \right.$$

Method:6 ➡ Tangent Plane and Normal Line



Tangent plane & normal line to the given surface at given point

→ The equation of the **tangent plane** to the surface $f(x, y) = 0$ at the point $P(x_1, y_1)$ is given by following formula.

$$\rightarrow (x - x_1) \left(\frac{\partial f}{\partial x} \right)_P + (y - y_1) \left(\frac{\partial f}{\partial y} \right)_P = 0$$

→ The equation of the **normal line** to the surface $f(x, y) = 0$ at the point $P(x_1, y_1)$ is given by following formula.

$$\rightarrow \frac{(x - x_1)}{\left(\frac{\partial f}{\partial x} \right)_P} = \frac{(y - y_1)}{\left(\frac{\partial f}{\partial y} \right)_P}$$

Tangent plane & normal line to the given surface at given point

→ The equation of the **tangent plane** to the surface $f(x, y, z) = 0$ at the point $P(x_1, y_1, z_1)$ is given by following formula.

$$\rightarrow (x - x_1) \left(\frac{\partial f}{\partial x} \right)_P + (y - y_1) \left(\frac{\partial f}{\partial y} \right)_P + (z - z_1) \left(\frac{\partial f}{\partial z} \right)_P = 0$$

→ The equation of the **normal line** to the surface $f(x, y, z) = 0$ at the point $P(x_1, y_1, z_1)$ is given by following formula.

$$\rightarrow \frac{(x - x_1)}{\left(\frac{\partial f}{\partial x} \right)_P} = \frac{(y - y_1)}{\left(\frac{\partial f}{\partial y} \right)_P} = \frac{(z - z_1)}{\left(\frac{\partial f}{\partial z} \right)_P}$$

Example of Tangent Plane and Normal Line

Example 1: Find the equation of **tangent plane** and **normal line** to the surface

$$2xz^2 - 3xy - 4x = 7 \text{ at point } P(1, -1, 2).$$

Solution: Here, $f(x, y, z) = 2xz^2 - 3xy - 4x - 7$.

$$\rightarrow \frac{\partial f}{\partial x} = 2z^2 - 3y - 4 \quad \Rightarrow \left(\frac{\partial f}{\partial x} \right)_{P(1, -1, 2)} = 2(2)^2 - 3(-1) - 4 = 7$$

$$\rightarrow \frac{\partial f}{\partial y} = -3x \quad \Rightarrow \left(\frac{\partial f}{\partial y} \right)_{P(1, -1, 2)} = -3(1) = -3$$

$$\rightarrow \frac{\partial f}{\partial z} = 4xz \quad \Rightarrow \left(\frac{\partial f}{\partial z} \right)_{P(1, -1, 2)} = 4(1)(2) = 8$$

Example of Tangent Plane and Normal Line

Solution Continue:

→ The equation of the **tangent plane** to the surface $f(x, y, z) = 0$ at the point $P(x_1, y_1, z_1) = P(1, -1, 2)$ is given by following formula.

$$\rightarrow (x - x_1) \left(\frac{\partial f}{\partial x} \right)_P + (y - y_1) \left(\frac{\partial f}{\partial y} \right)_P + (z - z_1) \left(\frac{\partial f}{\partial z} \right)_P = 0$$

$$\Rightarrow (x - 1)(7) + (y + 1)(-3) + (z - 2)(8) = 0$$

$$\Rightarrow 7x - 7 - 3y - 3 + 8z - 16 = 0$$

$$\Rightarrow \boxed{7x - 3y + 8z - 26 = 0}$$

Example of Tangent Plane and Normal Line

Solution Continue:

→ The equation of the **normal line** to the surface $f(x, y, z) = 0$ at the point $P(x_1, y_1, z_1) = P(1, -1, 2)$ is given by following formula.

$$\rightarrow \frac{(x - x_1)}{\left(\frac{\partial f}{\partial x}\right)_P} = \frac{(y - y_1)}{\left(\frac{\partial f}{\partial y}\right)_P} = \frac{(z - z_1)}{\left(\frac{\partial f}{\partial z}\right)_P}$$

$$\Rightarrow \boxed{\frac{x - 1}{7} = \frac{y + 1}{-3} = \frac{z - 2}{8}}$$

Example of Tangent Plane and Normal Line

Example 8: Show that the surfaces $z = xy - 2$ and $x^2 + y^2 + z^2 = 3$ have the same tangent plane at point $P(1, 1, -1)$.

Solution: Let, $f_1(x, y, z) = z - xy + 2$.

$$\rightarrow \frac{\partial f_1}{\partial x} = -y$$

$$\rightarrow \frac{\partial f_1}{\partial y} = -x$$

$$\rightarrow \frac{\partial f_1}{\partial z} = 1$$

$$\Rightarrow \left(\frac{\partial f_1}{\partial x} \right)_{P(1,1,-1)} = -1$$

$$\Rightarrow \left(\frac{\partial f_1}{\partial y} \right)_{P(1,1,-1)} = -1$$

$$\Rightarrow \left(\frac{\partial f_1}{\partial z} \right)_{P(1,1,-1)} = 1$$

Example of Tangent Plane and Normal Line

Solution Continue:

→ The equation of the **tangent plane** to the surface $f_1(x, y, z) = 0$ at the point $P(x_1, y_1, z_1) = P(1, 1, -1)$ is given by following formula.

$$\rightarrow (x - x_1) \left(\frac{\partial f_1}{\partial x} \right)_P + (y - y_1) \left(\frac{\partial f_1}{\partial y} \right)_P + (z - z_1) \left(\frac{\partial f_1}{\partial z} \right)_P = 0$$

$$\Rightarrow (x - 1)(-1) + (y - 1)(-1) + (z + 1)(1) = 0$$

$$\Rightarrow -x + 1 - y + 1 + z + 1 = 0$$

$$\Rightarrow -x - y + z + 3 = 0$$

$$\Rightarrow \boxed{x + y - z - 3 = 0}$$

Example of Tangent Plane and Normal Line

Example 8: Show that the surfaces $z = xy - 2$ and $x^2 + y^2 + z^2 = 3$ have the same tangent plane at point $P(1, 1, -1)$.

Solution Continue: Let, $f_2(x, y, z) = x^2 + y^2 + z^2 - 2$.

$$\rightarrow \frac{\partial f_2}{\partial x} = 2x$$

$$\rightarrow \frac{\partial f_2}{\partial y} = 2y$$

$$\rightarrow \frac{\partial f_2}{\partial z} = 2z$$

$$\Rightarrow \left(\frac{\partial f_1}{\partial x} \right)_{P(1,1,-1)} = 2$$

$$\Rightarrow \left(\frac{\partial f_1}{\partial y} \right)_{P(1,1,-1)} = 2$$

$$\Rightarrow \left(\frac{\partial f_1}{\partial z} \right)_{P(1,1,-1)} = -2$$

Example of Tangent Plane and Normal Line

Solution Continue:

→ The equation of the **tangent plane** to the surface $f_2(x, y, z) = 0$ at the point $P(x_1, y_1, z_1) = P(1, 1, -1)$ is given by following formula.

$$\rightarrow (x - x_1) \left(\frac{\partial f_2}{\partial x} \right)_P + (y - y_1) \left(\frac{\partial f_2}{\partial y} \right)_P + (z - z_1) \left(\frac{\partial f_2}{\partial z} \right)_P = 0$$

$$\Rightarrow (x - 1)(2) + (y - 1)(2) + (z + 1)(-2) = 0$$

$$\Rightarrow 2x - 2 + 2y - 2 - 2z - 2 = 0$$

$$\Rightarrow 2x + 2y - 2z - 6 = 0$$

$$\Rightarrow \boxed{x + y - z - 3 = 0}$$

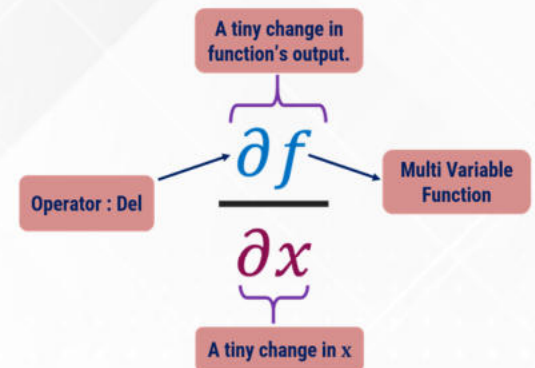
Hence, both the surfaces $z = xy - 2$ and $x^2 + y^2 + z^2 = 3$ have the **same tangent plane** at point $P(1, 1, -1)$.

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**THANK
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